FaCTS: Fact-Conditioned Generation and Sentence-Level Verification for Faithful Clinical Note Generation from Doctor–Patient Conversations

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Abstract—Large language models (LLMs) can draft clinical notes from doctor–patient conversations, but a fluent note may contain symptoms, medications or doses that were never said. We study this problem on two public benchmarks, ACI-Bench (full visit notes) and MTS-Dialog (note sections), and propose FaCTS, a pipeline that (i) extracts clinical facts with negation and dosage attributes from the transcript, (ii) conditions an open 8B LLM (Llama-3.1-8B-Instruct, 4-bit) on a fact table, and (iii) checks every generated sentence against the best-matching transcript turns with a BM25 retriever and a DeBERTa-v3 natural-language-inference verifier, dropping unsupported sentences. A plain QLoRA fine-tuned baseline is trained for comparison. On ACI-Bench test 1 the untrained model reaches ROUGE-1 0.519, equal to GPT-4 zero-shot (0.518) in the MEDIQA-Chat 2023 shared task, against 0.614 for the best GPT-4 in-context system; on MTS-Dialog the fine-tuned model reaches 0.400, level with fine-tuned shared-task systems (0.395–0.401). However, QLoRA fine-tuning raised the entity-level hallucination rate from 14.4% to 51.7% (ACI-Bench) and from 20.4% to 35.6% (MTS-Dialog); giving the fine-tuned model the fact table did not undo this. The fact table lowered hallucination to 10.6% and 15.8% without loss of coverage, giving the best fact-level F1 (82.8 and 77.4); verification lowered it further to 8.7% and 14.2% at a cost in coverage.

Index Terms—Clinical note generation, hallucination, large language models, QLoRA, natural language inference, factual consistency, doctor–patient dialogue summarization.

I. Introduction

A. Problem domain

Clinicians spend a large share of their working time on documentation; physicians can spend up to two hours on the electronic health record for every hour of direct patient care [1]. Automatically drafting the visit note from the recorded consultation — ambient clinical documentation — is therefore one of the most requested applications of LLMs in healthcare [2], [3]. Public benchmarks for the task now exist: ACI-Bench pairs full doctor–patient encounters with complete visit notes [3], and MTS-Dialog pairs shorter conversations with individual note sections [4]. The MEDIQA-Chat 2023 shared task used both and showed that LLM-based systems reach the highest automatic scores [5].

Reference draft for the team (AI-assisted). It shows the structure, content and evidence the Review 3 paper needs. Rewrite it in your own words before submission. Every experimental number is reproducible from the repository (reports/eval_review3/).

B. Problem statement

High ROUGE or BERTScore does not mean that a note is safe. A clinical note is a legal and medical record: a single invented allergy, a flipped negation ("denies chest pain" → "reports chest pain") or a changed dose can lead to a wrong treatment decision [2]. LLMs produce such hallucinations — content not supported by the source — even when the text reads fluently [6], [7], and general-domain hallucination detectors transfer poorly to clinical dialogue [6]. A second, less discussed problem concerns how open models are adapted: supervised fine-tuning (SFT) on targets that contain information not present in the input has been shown to increase hallucination in general-domain LLMs [8], [9]. Clinical reference notes contain exactly such information (content the clinician knew but did not say aloud), so standard fine-tuning may teach a model to write plausible but ungrounded details.

Formally, given a de-identified transcript D of a completed outpatient consultation, the system must produce a structured note $\hat{y}$ (a full note for ACI-Bench, one requested section for MTS-Dialog) such that every clinical statement in $\hat{y}$ is supported by D, while covering the clinically important facts of the reference note y. Speech recognition, live deployment and EHR integration are out of scope.

C. Existing approaches

Systems in the literature fall into three groups: (a) fine-tuned encoder–decoder models such as BART and LongT5 [4], [10]; (b) in-context learning with large proprietary models such as GPT-4 [5], [11]; and (c) parameter-efficient adaptation of open LLMs with QLoRA or prompt tuning [1], [12]. Most report lexical overlap only; factuality is either not measured or measured by human review on small samples [1], [11].

D. Proposed idea

Instead of trusting the generator, FaCTS constrains it from both sides. Before generation, a deterministic clinical fact extractor turns the transcript into a fact table (entity,

category, negation status, dose) that is placed in the prompt. After generation, each sentence is verified against the transcript by an NLI model and unsupported sentences are removed. Both stages work with any generator and need no GPU training.

E. Contributions

- A reproducible evaluation of an open 8B LLM on the official ACI-Bench and MTS-Dialog test sets, compared with the published MEDIQA-Chat 2023 systems using the official ROUGE configuration.
- Evidence that plain QLoRA fine-tuning raises the entity-level hallucination rate compared with the same model used zero-shot, consistent with findings on fine-tuning and new knowledge [8].
- FaCTS: a fact-conditioned prompt built from a negation-aware clinical fact extractor, and a sentence-level BM25 + NLI verifier with flag-and-drop, evaluated as an ablation (zero-shot → +fact table → +verifier).
- A clinical factuality evaluation that reports hallucination, negation-flip, numerical consistency and coverage, and maps them to fact-level precision, recall and F1.

F. Organisation

Section II reviews fifteen related papers and states the research gap. Section III describes the proposed system, the LLM architecture, the equations and the algorithm. Section IV presents the datasets, tools, metrics and results on each dataset, with comparison to published results. Section V concludes and outlines future work.

II. Literature Survey

A. Papers explaining the problem

Asgari et al. [2] propose CREOLA, a clinician-in-the-loop framework with an error taxonomy and a clinical-safety grading for LLM-generated notes. Over 18 experimental configurations and 12,999 clinician-annotated sentences they measure a 1.47% hallucination rate and a 3.45% omission rate, and show that prompt and workflow changes can push major errors below reported human error rates. The work establishes that hallucinations and omissions must be measured separately and graded by harm.

BN et al. [6] study hallucinations in summaries of patient–clinician dialogues. They build a fact-controlled "leave-N-out" dataset by deleting facts from source dialogues, and a set of natural hallucinations, and find that general-domain detectors fail on clinical hallucinations and that performance on synthetic hallucinations does not predict performance on natural ones; fact-counting detectors give explainable scores.

Das et al. [7] evaluate open-source LLMs (e.g., Qwen2.5, DeepSeek-v2) on extracting key events from discharge summaries and categorise the hallucinations they produce. The models capture admission reasons well but are inconsistent on follow-up recommendations, showing that errors concentrate in specific clinical content types.

Gekhman et al. [8] show in a controlled study that fine-tuning examples which introduce knowledge new to the model are learned slowly, and that once learned they linearly increase the model's tendency to hallucinate. Kaplan et al. [9] confirm that exposure to new facts through SFT increases hallucination with respect to pre-trained knowledge, trace it to interference between overlapping representations, and mitigate it with self-distillation or parameter freezing. Both are general-domain studies; they explain why fine-tuning on clinical notes that contain unspoken information can raise hallucination.

B. Papers on existing approaches

Ben Abacha et al. [5] organised MEDIQA-Chat 2023: Task A summarises MTS-Dialog conversations into note sections (1,201/100/200 train/validation/test) and Task B generates full ACI-Bench notes (67/20/40). Short notes were ranked by the mean of ROUGE-1, BERTScore and BLEURT, full notes by ROUGE-1; GPT-4 in-context learning ranked first in both tasks.

Giorgi et al. (WangLab) [11] compare fine-tuning a Longformer encoder–decoder with few-shot in-context learning of GPT-4 using retrieved similar examples; the ICL system ranked first (ROUGE-1 0.614 on ACI-Bench test 1), and physicians preferred its notes about as often as the human-written ones.

Sharma et al. (Cadence) [10] take a data-augmentation-first approach: a BART model generates synthetic dialogues from MIMIC-IV notes (Task C), which are added to the training data of BART summarisers with an N-pass strategy for long inputs; theirs were the only stable and reproducible runs on all three tasks.

Van Veen et al. [1] adapt eight LLMs with in-context learning and QLoRA across four clinical summarisation tasks, including doctor–patient dialogue (ACI-Bench). In a reader study with ten physicians, summaries of the best adapted model were non-inferior to human summaries in 45% of cases and preferred in 36%; they also report fabricated content in both LLM and human summaries.

Lyu et al. [12] apply soft prompt tuning to GatorTronGPT (5B and 20B) on MTS-Dialog. The 20B model with 302M trainable parameters reached ROUGE-1 0.363 on the 400-item test set, above a fully fine-tuned T5-Large (0.343), showing that parameter-efficient adaptation of large decoders is competitive.

C. Papers related to our methodology

Dettmers et al. [13] introduce QLoRA: back-propagation through a frozen 4-bit NormalFloat (NF4) model into low-rank adapters, with double quantisation and paged optimisers, which matches 16-bit fine-tuning while fitting large models on one GPU. Our baseline uses QLoRA unchanged.

Hu et al. [14] propose LoRA, which freezes pre-trained weights and learns a low-rank update $\Delta W = BA$ in each adapted layer, cutting trainable parameters by orders of magnitude without inference latency.

Laban et al. [15] show that NLI models detect summary inconsistencies well when applied at sentence level rather than document level (SummaC). Our verifier follows this idea, but retrieves the relevant transcript turns first because a dialogue is too long and too noisy to serve as a premise.

Huang et al. (SWING) [16] use NLI signals during training of dialogue summarisers to increase coverage of reference content while penalising inconsistent sentences, demonstrating that NLI judgements transfer to conversational input.

Seethakantha et al. [17] use hallucination detectors to guide iterative revision of clinical summaries (HDSR) and turn the revision trajectories into preference pairs (HDSR-PL), reducing hallucinations of Llama-3.1-8B-Instruct by 24% and 48% respectively on MIMIC-IV notes while preserving fluency. In the same study, supervised fine-tuning on clinician-written references nearly doubled the clinician-annotated hallucination count of Llama-3.1-8B-Instruct compared with zero-shot prompting (57 vs. 29), the clinical counterpart of [8]. Detector-guided repair and preference learning are the two directions our verifier and our planned preference-tuning stage follow.

D. Comparison and research gap

Table I compares the fifteen papers. Three gaps emerge. (1) Systems on the public dialogue benchmarks [5], [10]–[12] report lexical metrics only; factuality is measured either on proprietary data [2] or on other tasks such as discharge summaries [7], [17]. (2) Parameter-efficient fine-tuning is used as the default way to adapt open models [1], [12], although fine-tuning can increase hallucination in general-domain QA [8], [9] and in discharge-note summarisation [17]; this has not been measured on the public doctor–patient dialogue benchmarks. (3) NLI verification works at sentence level on news [15] and chit-chat dialogue [16], but clinical dialogues need evidence retrieval and negation handling first. FaCTS addresses (1) by reporting entity-level factuality next to the official ROUGE on ACI-Bench and MTS-Dialog, (2) by measuring zero-shot vs QLoRA hallucination on the same test sets, and (3) by combining a negation-aware fact table with retrieval-then-NLI verification.

III. Proposed Methodology

A. Technology

FaCTS is implemented in Python with PyTorch, Hugging Face Transformers, PEFT and bitsandbytes for the LLM, rank-bm25 for retrieval and a DeBERTa-v3 cross-encoder for NLI. The generator is Llama-3.1-8B-Instruct [18] quantised to 4-bit NF4. Training and generation run on Kaggle (2×NVIDIA T4, 16 GB each); fact extraction, verification and evaluation run on a laptop (NVIDIA RTX 2050, 4 GB).

B. System architecture

Fig. 1 shows the pipeline. It has four stages: (1) clinical fact extraction, (2) fact-table construction and prompt building, (3) LLM generation, and (4) evidence retrieval, NLI verification and filtering. The output is the clinical note, a verdict (green/yellow/red) for each sentence, and the ids of the transcript turns that support it (provenance).

C. Stage 1: clinical fact extraction

The clinical fact extractor (CFE) segments the transcript into speaker turns and sentences, matches lexicons of symptoms, conditions, medications and procedures, and attaches attributes. Each fact is a tuple

$$f = (\text{cat}, \text{text}, \text{canonical}, \text{polarity}, \text{certainty}, \text{temporality}, \text{experiencer}, \text{value}, \text{unit}), \quad (1)$$

over eight categories (symptom, medical history, diagnosis, medication, dosage, investigation, treatment/follow-up, numeric). Negation uses NegEx-style trigger windows [20] and a dialogue rule: a patient's "no" answers the clinical terms of the preceding doctor question. On a 40-fact manual sample the extractor reached 87.5% fully correct facts.

D. Stage 2: fact table and prompt

Mentions of the same canonical entity are merged. The status of an entity e with mentions M_e is taken from the mention that wins under the order: certain before uncertain, patient before doctor, later before earlier:

$$\text{status}(e) = \text{status}(m_e^*), \quad m_e^* = \arg\max_{m \in M_e} \big(c_m, p_m, t_m\big), \quad (2)$$

where $c_m = 1$ if mention m is certain, $p_m = 1$ if it was said by the patient, t_m is its position in the transcript, and tuples are compared lexicographically. Entities are listed by category with tags (DENIED), (POSSIBLE), (FAMILY), (PAST); doses are folded into their drug. The prompt is the Review 2 instruction, the rule "use these facts, keep (DENIED) facts negative, never add facts not in the transcript", the fact table and the full transcript. The table is about 15–20% of the transcript length. The transcript stays in the prompt because the lexicon can miss facts.

E. Stage 3: the LLM — architecture and attention

Llama-3.1-8B-Instruct [18] is a decoder-only Transformer [19] with 32 layers, model dimension $d = 4096$, 32 query heads and 8 key/value heads (grouped-query attention, head dimension $d_h = 128$), a SwiGLU feed-forward

TABLE I
Comparison of the surveyed papers

Paper	Year	Data	Method	Key result	Limitation for our problem
Problem					
Asgari et al. [2]	2025	Proprietary consultations	CREOLA error taxonomy, clinician review	1.47% hallucination, 3.45% omission (sentences)	Manual; not on public benchmarks
BN et al. [6]	2025	Patient–clinician dialogue summaries	Fact-controlled leave-N-out hallucinations	General detectors fail on clinical text	Detection only, no mitigation
Das et al. [7]	2025	Discharge summaries	Open LLMs, key-event extraction	Follow-up actions least reliable	Discharge notes, not dialogue
Gekhman et al. [8]	2024	Closed-book QA	Controlled fine-tuning on new knowledge	New knowledge raises hallucination linearly	General domain
Kaplan et al. [9]	2026	QA, synthetic facts	Self-distillation SFT, freezing	Reduces SFT-induced hallucination	General domain
Existing approaches					
Ben Abacha et al. [5]	2023	MTS-Dialog, ACI-Bench	Shared task, 14 (A) and 9 (B) teams	GPT-4 ICL best	Lexical/semantic metrics only
Giorgi et al. [11]	2023	ACI-Bench, MTS-Dialog	GPT-4 ICL vs fine-tuned LED	R-1 0.614 (ACI test 1)	Proprietary model; no factuality metric
Sharma et al. [10]	2023	MTS-Dialog, ACI-Bench	BART + synthetic dialogues	Stable, reproducible runs	Small model; no factuality metric
Van Veen et al. [1]	2024	6 datasets incl. ACI-Bench	ICL vs QLoRA, 8 LLMs, reader study	LLM preferred/non-inferior in 81%	A&P section only for dialogue
Lyu et al. [12]	2024	MTS-Dialog	Soft prompt tuning, GatorTronGPT-20B	R-1 0.363 (400 test)	20B model, 8×A100; lexical only
Methodology					
Dettmers et al. [13]	2023	Instruction data	QLoRA: NF4, double quant., paged opt.	16-bit quality on one GPU	No factuality control
Hu et al. [14]	2022	GLUE, generation	Low-rank adapters	Large reduction in trainable params	No factuality control
Laban et al. [15]	2022	SummaC benchmark (news)	Sentence-level NLI	Strong inconsistency detection	News, not dialogue
Huang et al. [16]	2023	DialogSum, SAMSum	NLI-based training signals	Better coverage and faithfulness	Chit-chat, not clinical
Seethakantha et al. [17]	2026	MIMIC-IV notes	Detector-guided refinement + preference learning	−48% hallucination; SFT doubles it (Llama-3.1-8B)	Notes, not dialogue; clinician labels needed

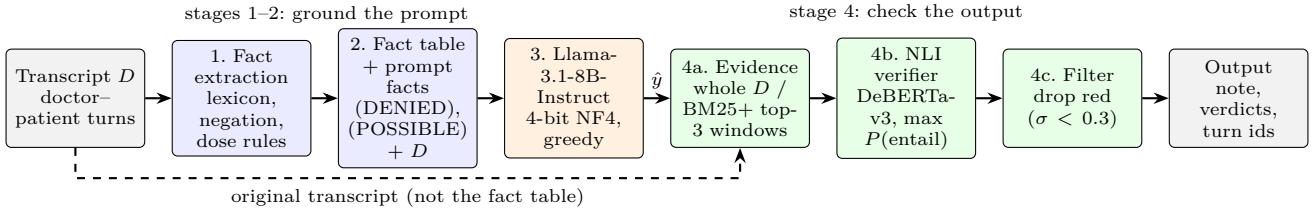


Fig. 1. FaCTS architecture. Stages 1–2 build a fact-conditioned prompt, stage 3 generates the note, and stage 4 verifies every sentence against the original transcript.

layer of width 14,336, RMSNorm pre-normalisation, rotary position embeddings (RoPE) and a 128,000-token vocabulary (Table 3 of [18]). Each layer computes

$$h' = h + \mathrm{MHA}\big(\mathrm{RMSNorm}(h)\big), \tag{3}$$

$$h_{\mathrm{out}} = h' + \mathrm{FFN}\big(\mathrm{RMSNorm}(h')\big). \tag{4}$$

For head i with query projection W_i^Q and the shared key/value projections $W_{g(i)}^K, W_{g(i)}^V$ of its group $g(i)$, causal self-attention is

$$\mathrm{head}_i = \mathrm{softmax}\Big(\frac{\tilde{Q}_i \tilde{K}_{g(i)}^\top}{\sqrt{d_h}} + M\Big) V_{g(i)}, \tag{5}$$

where $\tilde{Q}, \tilde{K}$ are the queries and keys after rotary embedding, and the causal mask $M_{tj} = 0$ if $j \le t$ and $-\infty$ otherwise, so token t attends only to the prompt and earlier generated tokens. Rotary embedding rotates each pair of dimensions $(2k, 2k{+}1)$ of a vector at position t by the angle $t\,\theta_k$, $\theta_k = b^{-2k/d_h}$ with base $b = 500{,}000$, so that $\tilde{q}_t^\top \tilde{k}_j$ depends only on the relative offset $t - j$.

The heads are concatenated and projected, $\mathrm{MHA}(x) = [\mathrm{head}_1; \ldots; \mathrm{head}_{32}]W^O$, and the feed-forward layer is

$$\mathrm{FFN}(x) = W_2\big(\mathrm{SiLU}(W_1 x) \odot W_3 x\big). \tag{6}$$

Generation is autoregressive and greedy: $\hat{y}_t = \arg\max_v p_\theta(v \mid \text{prompt}, \hat{y}_{<t})$, with a repetition penalty of 1.1 and at most 1,024 new tokens. The weights are stored in 4-bit NF4 with blockwise absmax scaling, $W \approx c_b\,\mathrm{NF4}(W_b/c_b)$ for each block b of 64 weights [13].

F. Baseline: QLoRA supervised fine-tuning

For comparison, the same model is fine-tuned with QLoRA [13]. LoRA [14] keeps the frozen weight W_0 and learns a low-rank update in every attention and feed-forward projection:

$$h = W_0 x + \frac{\alpha}{r} BAx, \quad B \in \mathbb{R}^{d\times r},\ A \in \mathbb{R}^{r\times k}, \tag{7}$$

with $r = 16$, $\alpha = 32$ (41.9M trainable parameters, 0.52% of 8.07B). The loss is the token-level cross-entropy on the note tokens only (prompt tokens are masked):

$$\mathcal{L}_{\mathrm{SFT}}(\theta) = -\frac{1}{|y|}\sum_{t=1}^{|y|} \log p_\theta\big(y_t \mid \text{prompt}, y_{<t}\big). \tag{8}$$

G. Stage 4: retrieval and NLI verification

The note is split into sentences; headers and fragments shorter than three words are kept but not checked. Spoken numbers and units in the transcript are first written as digits ("forty M G" → "40 mg"), because the NLI model does not equate the two. The whole transcript is one candidate premise when it fits the NLI context (at most 400 tokens, true for most MTS-Dialog items). In addition, the transcript is split into windows w_j = turn j plus turn $j-1$, so that a short answer ("No.") keeps the question it answers, and each window is scored with BM25+ [21]:

$$\mathrm{BM25}^{+}(s, w) = \sum_{q\in s} \mathrm{IDF}(q) \times \Big(\frac{f(q,w)\,(k_1+1)}{f(q,w) + k_1\big(1 - b + b\frac{|w|}{\overline{|w|}}\big)} + \delta\Big), \tag{9}$$

with $k_1 = 1.5$, $b = 0.75$, $\delta = 1$. (Plain BM25 assigns zero or negative IDF to most terms when a transcript has only a few turns, and then returns arbitrary windows.) A DeBERTa-v3 [22] cross-encoder trained on SNLI/MultiNLI scores each candidate premise E in the set $\mathcal{E}_s$ = {whole transcript (if it fits), each of the top-3 windows, the three windows joined}, and the support score is the best entailment probability:

$$\sigma(s) = \max_{E\in\mathcal{E}_s} \mathrm{softmax}\big(f_{\mathrm{NLI}}([E; s])\big)_{\mathrm{entail}}. \tag{10}$$

This "max" strategy was chosen on the validation sets only (Section IV-H), never on test outputs. Sentences are labelled green ($\sigma \geq 0.6$), yellow ($0.3 \leq \sigma < 0.6$) or red ($\sigma < 0.3$). In the evaluated configuration red sentences are dropped (flag-and-drop); rewriting them with the LLM is left as an option.

Algorithm 1 FaCTS note generation

Require: transcript D, task (full note or section c), generator G, NLI model f

$F \leftarrow \mathrm{ExtractFacts}(D)$ ▷ lexicon, negation, doses
$T \leftarrow \mathrm{FactTable}(F)$ ▷ merge mentions, resolve status
$P \leftarrow \mathrm{Prompt}(\text{task}, c, T, D)$
$\hat{y} \leftarrow G.\mathrm{GreedyDecode}(P)$
$D' \leftarrow \mathrm{DigitiseNumbers}(D)$
$W \leftarrow \{\,\mathrm{turn}_{j-1} \oplus \mathrm{turn}_j\,\}_j$ of D'; build BM25+ index over W
$\hat{y}' \leftarrow$ empty note
for each piece s of $\mathrm{SplitSentences}(\hat{y})$ do
  if s is a header or has < 3 words then append s to $\hat{y}'$; continue
  end if
  $B \leftarrow$ top-3 windows of W by $\mathrm{BM25}^{+}(s, \cdot)$
  $\mathcal{E}_s \leftarrow B \cup \{\oplus B\} \cup (\{D'\}$ if $|D'| \leq 400$ tokens$)$
  $\sigma \leftarrow \max_{E\in\mathcal{E}_s} f(E, s)_{\mathrm{entail}}$
  label $\leftarrow$ green if $\sigma \geq 0.6$, yellow if $\sigma \geq 0.3$, else red
  if label $\neq$ red then append s to $\hat{y}'$
  end if
  record $(s, \mathrm{ids}(E_s), \sigma, \text{label})$
end for
return $\hat{y}'$ and the verdict records

TABLE II
Datasets and splits used

Corpus	Train	Val.	Test	Target
ACI-Bench (Dataset 1)	67	20	120 (3×40)	full visit note
MTS-Dialog (Dataset 2)	1,198	100	400 (2×200)	one note section
DialogueSOAP (train)	3,000	100	–	SOAP (synthetic)
SFT training mix	4,265	220	–	

H. Algorithm

I. Outputs

For every encounter the system returns the note, a per-sentence verdict with its supporting turn ids, and the fact table, so a clinician can see why each sentence was kept and which turns support it.

IV. Experimentation and Results

A. Datasets

Dataset 1: ACI-Bench [3]: role-played and real-style visit conversations with full notes (median 404 words). We use the three official test sets (test 1 = MEDIQA-Chat Task B, test 2, test 3; 40 each). Dataset 2: MTS-Dialog [4]: short conversations with one note section (median 14 words) and its section header. We use test 1 (200, MEDIQA-Chat Task A) and test 2 (200, MEDIQA-Sum 2023). The SFT baseline was trained on the official training sets of both corpora plus 3,000 synthetic DialogueSOAP pairs [23] (4,265 training, 220 validation examples); no test item was used for training or tuning.

B. Systems compared

- SFT (QLoRA): Review 2 adapter, 332 steps (1.24 epochs), best validation loss at step 200.
- SFT + Fact table: the same adapter with the fact-conditioned prompt of B.
- A. Zero-shot: base Llama-3.1-8B-Instruct, same prompt as SFT.
- B. +Fact table: A with the fact-conditioned prompt (FaCTS stages 1–3).
- C. +Verifier: B followed by retrieval + NLI flag-and-drop (full FaCTS).

All systems use greedy decoding with repetition penalty 1.1 and at most 1,024 new tokens, on the same 520 test items.

C. Software tools

PyTorch 2.8–2.10, Transformers 4.57, PEFT 0.19, bitsandbytes 0.50, Liger-Kernel (training), rank-bm25, rouge-score, bert-score, Kaggle (2×T4) for the LLM, and a laptop RTX 2050 for verification and evaluation.

D. Evaluation metrics

Metrics used by the surveyed papers. ROUGE-1/2/L/Lsum F1 [24] computed exactly as the MEDIQA-Chat 2023 evaluation script (Hugging Face evaluate, no stemming, mean over items), so our numbers are directly comparable with Tables III and IV. BERTScore [25] (roberta-large, rescaled) is reported between our systems only, because the shared task used a different model (deberta-xlarge-mnli) without rescaling.

Clinical factuality. Facts are extracted with the CFE from the transcript, the reference and the prediction. With $\mathcal{P}$ the predicted entities, $\mathcal{S}$ the transcript entities and $\mathcal{R}$ the reference entities:

$$\text{Halluc.} = \frac{|\mathcal{P} \setminus \mathcal{S}|}{|\mathcal{P}|}, \qquad \text{Coverage} = \frac{|\mathcal{R} \cap \mathcal{P}|}{|\mathcal{R}|}, \tag{11}$$

$$\text{Neg. flip} = \frac{|\{e \in \mathcal{R} \cap \mathcal{P} : \text{pol}_{\mathcal{P}}(e) \neq \text{pol}_{\mathcal{R}}(e)\}|}{|\mathcal{R} \cap \mathcal{P}|}, \tag{12}$$

$$\text{Num. cons.} = 1 - \frac{\#\text{numbers in } \hat{y} \text{ absent from } D \text{ and } y}{\#\text{numbers in } \hat{y}}. \tag{13}$$

Following the requirement to report precision/recall/F1, we define fact precision $P_f = 1 - \text{Halluc.}$ (share of predicted facts grounded in the transcript), fact recall $R_f = \text{Coverage}$ and $F1_f = 2P_fR_f/(P_f + R_f)$. "Accuracy" is not defined for free-text generation and is not reported. All rates are per-item means.

E. Implementation on Dataset 1: ACI-Bench

Table III compares our systems with the published MEDIQA-Chat 2023 Task B systems on ACI-Bench test 1, scored with the official script. The untrained Llama-3.1-8B-Instruct (A) reaches ROUGE-1 0.519, equal to the organisers' GPT-4 zero-shot baseline (0.518) and above ChatGPT (0.474), the fine-tuned LongT5 (0.431) and the GPT-3 system of GersteinLab (0.501). It stays below the best systems, which use GPT-4 with retrieved in-context examples (0.614) or fine-tuned BART (0.530 with two-pass summarisation [10]; 0.535 when generating each note division separately [3]). Adding the fact table (B) leaves ROUGE essentially unchanged (0.516). The verifier (C) lowers ROUGE-1 to 0.462 because it removes sentences. The QLoRA SFT model scores only 0.255: 89 of its 120 ACI-Bench notes reached the 1,024-token limit and 70 were degenerate (repeated phrases or endless lists such as "no history of nipple edema …"), because only 67 of its 4,265 training examples were full ACI-Bench notes. Giving it the fact table (SFT + Fact table) does not help (0.208): the fine-tuned model ignores the added context and loops in the same way. An earlier run with a 768-token limit scored 0.281; allowing more tokens only lengthened the loops, so the low score is caused by degeneration, not by the length limit.

TABLE III
Dataset 1 — ACI-Bench test 1 (40 full notes, MEDIQA-Chat 2023 Task B). ROUGE F1, official scoring. Published rows from [5].

System	Method	R-1	R-2	R-L	R-Lsum
WangLab [11]	GPT-4 ICL	0.614	0.329	0.382	0.551
Cadence [10]	BART FT	0.530	0.250	0.298	0.466
iuteam1 [5]	LSG-BART FT	0.527	0.262	0.306	0.498
GersteinLab [5]	GPT-3	0.501	0.251	0.328	0.467
Calvados [5]	LongT5 FT	0.431	0.202	0.239	0.386
Yim et al. [3]	BART+SAMSum, by division	0.535	0.251	–	–
Organisers [5]	GPT-4 0-shot	0.518	0.226	0.303	0.426
Organisers [5]	ChatGPT 0-shot	0.474	0.190	0.271	0.390
Ours: SFT (QLoRA)	Llama-3.1-8B	0.255	0.101	0.159	0.231
Ours: SFT + Fact table	Llama-3.1-8B	0.208	0.079	0.126	0.187
Ours: A. Zero-shot	Llama-3.1-8B	0.519	0.214	0.297	0.477
Ours: B. + Fact table	Llama-3.1-8B	0.516	0.214	0.295	0.474
Ours: C. + Verifier (FaCTS)	Llama-3.1-8B	0.462	0.195	0.277	0.427

F. Implementation on Dataset 2: MTS-Dialog

On MTS-Dialog (Table IV) the picture reverses. The SFT model reaches ROUGE-1 0.400 on test 1, level with GersteinLab (0.401) and Calvados (0.395) and above both organiser baselines (GPT-4 0.307, ChatGPT 0.303); on all 400 test items it reaches 0.392, above the prompt-tuned GatorTronGPT-20B (0.363) and the fully fine-tuned T5-Large (0.343) of [12]. The zero-shot model scores 0.303, the same as GPT-4 zero-shot. With the fact table the SFT model scores 0.382 (test 1) and 0.372 (all 400). MTS-Dialog references are very short (median 14 words); fine-tuning teaches the model this terse style, whereas the base model writes full sentences ("The patient presents with a complaint of a boil-like condition located on their buttocks" for the reference "Buttock abscess."), which lowers lexical overlap without being wrong. Our prompt contains the gold section header, whereas Task A teams also had to predict it.

G. Clinical factuality on both datasets

Table V reports the factuality metrics on all test items and Fig. 2 summarises them; Table VI gives paired differences with bootstrap confidence intervals.

TABLE IV
Dataset 2 — MTS-Dialog. Top: test 1 (200, MEDIQA-Chat 2023 Task A [5]). Bottom: both test sets (400) vs. [12]. ROUGE F1, official scoring.

System	Method	R-1	R-2	R-L
Test 1 (200)				
WangLab [11]	GPT-4 ICL	0.447	0.228	0.384
Cadence [10]	BART FT	0.430	0.208	0.364
GersteinLab [5]	FT + ICL	0.401	0.215	0.332
Calvados [5]	LongT5 FT	0.395	0.186	0.332
Care4Lang [5]	LM FT	0.358	0.165	0.289
Organisers [5]	GPT-4 0-shot	0.307	0.128	0.236
Organisers [5]	ChatGPT 0-shot	0.303	0.121	0.242
Ours: SFT (QLoRA)	Llama-3.1-8B	0.400	0.183	0.342
Ours: SFT + Fact table	Llama-3.1-8B	0.382	0.175	0.326
Ours: A. Zero-shot	Llama-3.1-8B	0.303	0.106	0.235
Ours: B. + Fact table	Llama-3.1-8B	0.294	0.117	0.227
Ours: C. + Verifier (FaCTS)	Llama-3.1-8B	0.287	0.114	0.225
Test 1 + 2 (400)				
Lyu et al. [12]	GatorTronGPT-20B PT	0.363	0.155	0.347
Lyu et al. [12]	T5-Large FT	0.343	0.146	0.337
Ours: SFT (QLoRA)	Llama-3.1-8B	0.392	0.168	0.333
Ours: SFT + Fact table	Llama-3.1-8B	0.371	0.164	0.313
Ours: A. Zero-shot	Llama-3.1-8B	0.310	0.108	0.240
Ours: B. + Fact table	Llama-3.1-8B	0.285	0.106	0.215
Ours: C. + Verifier (FaCTS)	Llama-3.1-8B	0.275	0.102	0.213

Fine-tuning increases hallucination. With the same prompt, the SFT model hallucinates 51.7% of its clinical entities on ACI-Bench and 35.6% on MTS-Dialog, against 14.4% and 20.4% for the untrained model (paired differences +37.3 and +14.3 points, both significant). On ACI-Bench it also flips more negations (+9.0), invents more numbers (numerical consistency −26.2) and covers fewer reference facts (−29.6). On MTS-Dialog, where only 12 of 400 SFT notes were truncated, the hallucination increase remains significant while ROUGE-L improves by 9.5 points: the model learned the form of the notes faster than their grounding. This agrees with the general-domain finding that fine-tuning on unknown facts increases hallucination [8] and with the doubled hallucination count reported for SFT of the same base model on discharge notes [17]. Adding the fact table to the SFT model lowers its hallucination on MTS-Dialog (35.6%→28.3%, −4.3 paired, significant) but not on ACI-Bench (51.2%, −0.6, not significant), and it increases negation flips on both datasets (+4.0 and +3.0). Even with the fact table the SFT model hallucinates about twice as often as the untrained model with the same prompt (B), so the fact table cannot repair what fine-tuning broke.

The fact table reduces hallucination without losing coverage. Conditioning on the extracted facts (B) lowers hallucination by 3.9 points on ACI-Bench (14.4%→10.6%) and by 3.5 points on MTS-Dialog (paired; 20.4%→15.8% over all items), both significant, and raises numerical consistency on ACI-Bench from 83.7% to 93.0%, because doses and measurements are listed explicitly. Coverage does not drop (77.2% vs. 76.8% on ACI-Bench; 71.5% vs. 68.0% on MTS-Dialog), so B has the best fact-level F1 on both datasets (82.8 and 77.4). Negation flips do not change significantly.

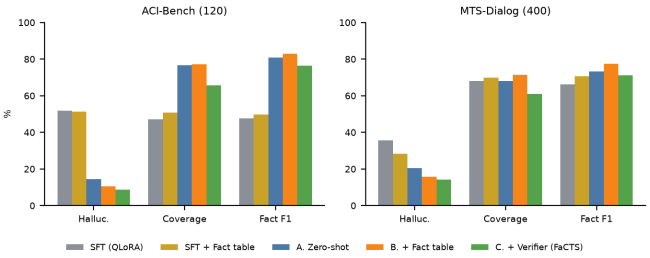


Fig. 2. Hallucination rate, coverage and fact-level F1 per system and dataset (all test items).

Drop-only verification trades coverage for precision. The verifier (C) gives the lowest hallucination rate on both datasets (8.7% and 14.2%), but the reduction over B is small (significant only on ACI-Bench, −1.8 points) while coverage falls by 11.6 and 10.6 points. Removing a sentence that contains one unsupported entity also removes the supported facts in it. FaCTS with the verifier is therefore a high-precision mode; the fact-conditioned generator alone gives the best balance.

H. Verifier analysis

The evidence strategy of the verifier was chosen on the validation sets only. Reference sentences of the 120 ACI-Bench and MTS-Dialog validation notes should pass; copies with one injected negation flip or changed number should fail. With the selected strategy (maximum over whole transcript, top-3 windows and their union), the verifier passed 72.8% of 966 reference sentences and caught 60.6% of 302 corrupted ones (75.4% of number changes but only 35.1% of negation flips), a balanced accuracy of 0.667, against 0.648 for a single retrieved premise and 0.639 with five windows. Because human references contain some content that was never said, the pass rate is a lower bound. On the test notes of system B the verifier marked 28.8% of the checked sentences red on both datasets (Table VII); no note was emptied. Its weakness on negation mirrors the finding that general-domain detectors struggle on clinical dialogue [6].

I. Discussion and limitations

Three results hold on both datasets with significant paired differences: QLoRA fine-tuning raised hallucination, the fact table lowered it, and the verifier lowered coverage. Lexical metrics alone would have ranked the SFT model best on MTS-Dialog, which is why factuality metrics must be reported next to ROUGE.

The study has limitations. (1) Factuality is measured automatically with our lexicon-based extractor; human reference notes themselves score 15% (ACI-Bench) and 28% (MTS-Dialog) “hallucination” under this metric, which is therefore a floor, not zero, and no clinician evaluation was performed. (2) The SFT adapter was

TABLE V
Clinical factuality on all test items of each dataset (per-item means, %). R-L uses stemming; BERTScore = roberta-large, rescaled. Halluc. and Neg. flip: lower is better. $P_f = 1-$Halluc., $R_f =$ Coverage.

System	R-L	BERTScore	Halluc.↓	Neg. flip↓	Num. cons.	Coverage	P_f	R_f	$F1_f$
ACI-Bench (120)									
SFT (QLoRA)	16.4	-0.022	51.7	13.7	57.6	47.2	48.3	47.2	47.7
SFT + Fact table	13.3	-0.102	51.2	17.7	62.6	50.7	48.8	50.7	49.7
A. Zero-shot	30.4	0.224	14.4	4.7	83.7	76.8	85.6	76.8	80.9
B. + Fact table	29.7	0.219	10.6	4.7	93.0	77.2	89.4	77.2	82.8
C. + Verifier (FaCTS)	27.8	0.181	8.7	3.8	93.6	65.6	91.3	65.6	76.3
MTS-Dialog (400)									
SFT (QLoRA)	34.6	0.428	35.6	5.1	82.9	68.1	64.4	68.1	66.2
SFT + Fact table	32.4	0.411	28.3	8.4	80.7	69.8	71.7	69.8	70.8
A. Zero-shot	25.1	0.223	20.4	4.9	88.3	68.0	79.6	68.0	73.3
B. + Fact table	22.4	0.224	15.8	5.4	84.4	71.5	84.2	71.5	77.4
C. + Verifier (FaCTS)	22.3	0.212	14.2	6.2	85.1	60.9	85.8	60.9	71.3

TABLE VI
Paired differences in percentage points (per-item means). *: 95% paired bootstrap interval excludes 0 (10,000 resamples).

Dataset	Comparison	Halluc.	Neg. flip	Num. cons.	Coverage	R-L
ACI-Bench	SFT − A	+37.3*	+9.0*	-26.2*	-29.6*	-14.0*
ACI-Bench	SFT+F − SFT	-0.6	+4.0*	+5.5	+3.5	-3.1*
ACI-Bench	B − A	-3.9*	-0.0	+9.4*	+0.4	-0.7*
ACI-Bench	C − B	-1.8*	-0.8	+0.8	-11.6*	-1.8*
MTS-Dialog	SFT − A	+14.3*	+1.2	-5.4*	+0.1	+9.5*
MTS-Dialog	SFT+F − SFT	-4.3*	+3.0*	-1.6	+1.7	-2.1*
MTS-Dialog	B − A	-3.5*	+1.2	-0.7	+3.6	-2.7*
MTS-Dialog	C − B	-0.7	+0.3	+2.3*	-10.6*	-0.1

TABLE VII
Verifier verdicts on system B notes (share of checked sentences, %). Red sentences are dropped.

Dataset	Notes	Sent.	Green	Yellow	Red	Emptied
ACI-Bench (120)	120	2905	68.0	3.2	28.8	0
MTS-Dialog (400)	400	1506	69.5	1.7	28.8	0

selected by validation loss on data with few full ACI-Bench notes; a rebalanced adapter might degenerate less, so its ACI-Bench numbers describe this adapter, not SFT in general. (3) BLEURT and the deberta-xlarge BERTScore of the shared task were not computed, so only ROUGE is compared with published systems. (4) MTS-Dialog prompts include the gold section header. (5) All results are from one greedy run.

V. Conclusion and Future Work

We evaluated an open 8B LLM for clinical note generation on the official ACI-Bench and MTS-Dialog test sets and compared it with published MEDIQA-Chat 2023 systems using the official ROUGE configuration. Untrained Llama-3.1-8B-Instruct matched GPT-4 zero-shot on both datasets (ROUGE-1 0.519 and 0.303). QLoRA fine-tuning raised ROUGE on the short MTS-Dialog sections to the level of fine-tuned shared-task systems (0.400), but increased the entity-level hallucination rate on both datasets (ACI-Bench 14.4%→51.7%, MTS-Dialog 20.4%→35.6%); adding the fact table to the fine-tuned model did not bring it back to the level of the untrained model. FaCTS addresses this without training: conditioning the model on a negation- and dose-aware fact table lowered hallucination to 10.6% and 15.8% with no loss of coverage, giving the best fact-level F1 on both datasets, and sentence-level NLI verification lowered it further to 8.7% and 14.2% at a cost of about 11 points of coverage.

Future work follows directly from these results. (1) Repair instead of drop: rewrite red sentences with the generator, as in detector-guided self-refinement [17], to keep their supported facts. (2) Factuality preference training: train an ORPO [26] adapter on pairs of reference notes and copies with one injected error (negation flip, dose change, entity swap, inserted fact), so that the model, not only the pipeline, learns to prefer grounded notes. (3) A better verifier for dialogue negation, e.g. fine-tuning the NLI model on clinical question–answer pairs. (4) Clinician evaluation of a sample of notes, and the shared-task BERTScore and BLEURT for a complete comparison. (5) Rebalancing the fine-tuning data towards full notes and selecting checkpoints by factuality rather than validation loss.

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